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UW24NoiseSrcDefTopic

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About the UltraWave24 MWNoiseSource

About the UltraWave24 MWNoiseSource
The MWNoiseSource is a wideband noise source that generates a calibrated excess noise ratio (ENR). The primary application of the MWNoiseSource is noise figure tests.
This section includes the following topics:

•	Features

- Safety Information
- Key Terms

MWNoiseSource information includes the following sections:

- UltraWave24 MWNoiseSource Theory of Operation
- UltraWave24 MWNoiseSource Programming
- UltraWave24 MWNoiseSource Debug Display

Related Topics

- UltraWave24 Specifications
- MWNoiseSource (VBT syntax)
- UltraWave24 DIB Design Guidelines
- About the UltraWave24 Subsystem
- UltraWave24 Examples

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About the UltraWave24 MWNoiseSource : Features

Features

The MWNoiseSource generates wideband noise at a calibrated ENR.

Basic functional specifications are:

- Frequency range: 50MHz to 6GHz
- Excess Noise Ratio (ENR) level: ~15dB (dependent on frequency)

For UltraWave24 hardware specifications, see UltraWave24 Specifications.

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About the UltraWave24 MWNoiseSource : Safety Information

Safety Information

When operating Microwave instruments, always follow general UltraFLEX test system safety guidelines. See Test System Safety.

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About the UltraWave24 MWNoiseSource : Key Terms

Key Terms

cold	Cold temperature condition simulated by a noise source switched off.
ENR	Excess noise ratio. The power of a noise source when it is on relative to when it is off.
hot	Hot temperature condition simulated by a noise source switched on.
noise figure	Noise added to the signal by a DUT.
target ENR frequency	Frequency of interest for target ENR.

For more MW terms and definitions, see Key Terms in About the UltraWave24 Subsystem.

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UltraWave24 MWNoiseSource Debug Display

UltraWave24 MWNoiseSource Debug Display

The UltraWave24 Microwave Debug Display can display one or more instrument blocks for the MWNoiseSource logical instrument, each showing the state of a MWNoiseSource in the context of a site.

This section includes the following topic:

- UltraWave24 MWNoiseSource Debug Display Controls

For an overview of all displays supporting the UltraWave24 subsystem, see UltraWave24 Displays.

For more information on using debug displays, see the following:

- Using Debug Displays in TDE
- Producing Code from Debug Displays

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UltraWave24 MWNoiseSource Debug Display : UltraWave24 MWNoiseSource Debug Display Controls

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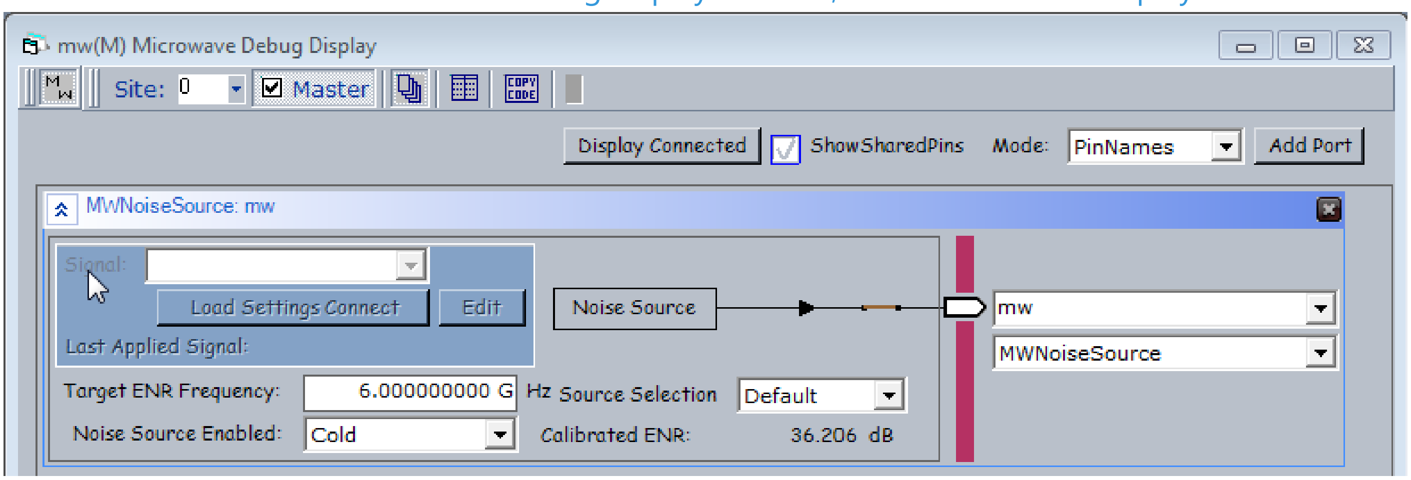
UltraWave24 MWNoiseSource Debug Display Controls

The display controls for the MWNoiseSource assigned to a port allow you to view and set some common MWNoiseSource parameters, which are also accessible through the Pins interface of the VBT language.

For information about the common debug display controls, see UltraWave24 Displays.



MWNoiseSource Logical Instrument Display

The following controls are available on the MWNoiseSource logical instrument block:

Target ENR Frequency	Displays and sets the target frequency for the noise measurement. Grayed out for UltraWave24 MWNoiseSource.
Noise Source Enabled	Displays and sets the noise source enabled mode: Cold or Hot.
Source Selection	Displays and sets the synthesizer used to generate noise source signals.
Calibrated ENR	Displays the calibrated ENR value for the target ENR and frequency.

Signal	Name of the signal used by the source. Available signals appear on the drop-down list. If a default signal has been defined, the default signal's name appears automatically in this field. If the menu contains no signal names, no signals were previously defined in VBT. The MWNoiseSource can operate without a signal.
Load Settings Connect	Loads the selected signal's timing settings as specified on the Mixed Signal Timing sheet, as well as all of the instrument's settings and connects the MWNoiseSource.
Edit	Brings up the PSet and Signals Editor (PSSE) with the selected signal loaded. This control is grayed out if no signals are available. See PSet and Signals Editor in the TDE section.
Info	Displays information for the selected signal in a pop-up balloon window.
Last Applied Signal	Name of the signal last loaded to the MWNoiseSource.
connection relay	The relay control connects or disconnects the MWNoiseSource instrument to the specified pin or channel. Click the relay to open or close the connection.

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UltraWave24 MWNoiseSource Programming

UltraWave24 MWNoiseSource Programming

The MWNoiseSource is a simple instrument with two key operations used to make a noise figure measurement of a DUT by the Y-factor method. The MWNoiseSource provides:

- A Hot noise power source
- A Cold noise power source

The following topic is available:

- MWNoiseSource Pattern Control

For general aspects of programming instruments in the UltraWave24 Subsystem, see UltraWave24 Instrument Programming.

For more information, see MWNoiseSource in the VBT Syntax Reference.

MWNoiseSource Reset Values

The table lists the local reset values for the MWNoiseSource.

MWNoiseSource Reset Values

Setting	Default Value
Enabled Temperature	Cold
TargetENRFrequency	1GHz
Connections	Disconnected

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UltraWave24 MWNoiseSource Programming : MWNoiseSource Pattern Control

MWNoiseSource Pattern Control

You can control an MWNoiseSource instrument using pattern microcodes, allowing changes to instrument states and parameters during a program. You can obtain repeatable timing by programming an instrument using pattern microcodes and Signals programming.

You create a pattern independently of the test program using the PatternTool or a text editor. Patterns are composed of a series of numbered digital vectors. These digital vectors define the data for each channel listed in the pattern, as well the pattern microcode used for controlling the instrument during pattern execution.

The following topics are available:

- MWNoiseSource Pattern Microcodes
- MWNoiseSource ASCII Pattern Syntax

☑ **Note:**

To use pattern microcodes to control an MWNoiseSource on UltraFLEX, you must include a digital pin in the pin map, the channel map, and the pin list of the pattern module.

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UltraWave24 MWNoiseSource Programming : MWNoiseSource Pattern Control : MWNoiseSource Pattern Microcodes

MWNoiseSource Pattern Microcodes

A microcode controls a single function, such a starting or stopping the sourcing of a signal. Extended microcode commands are available for use inside a pattern. Patterns allow you to control the precise timing of when a signal is sourced. To establish repeatable timing, or to synchronize with other instruments (such as the digital subsystem) in the tester, use a pattern. If you want to start generating a signal but are not concerned with precise timing, however, then you do not need a pattern.

The table lists the extended microcodes used to issue commands to an MWNoiseSource instrument from a pattern.

MWNoiseSource Pattern Microcodes

Microcode	Functional Description
Disconnect	Instructs an MWNoiseSource to reset all connections to their disconnected state.
LoadSettingsConnect <SignalName>	Instructs an MWNoiseSource to load the settings for the named capture signal and connects the hardware to the DUT. SignalName is required.

Note the following when using microcodes with an MWNoiseSource:

- IG-XL has rules regarding the necessary spacing in vectors between microcodes on a specific channel. If you place microcodes too close together, IG-XL will return an error. General guidelines for spacing microcodes include the following:

- LoadSettingsConnect: 400s (typical)
- Disconnect: <200s (typical)

- An MWNoiseSource does not use PSets. It provides the equivalent capability using Signals programming in VBT.
- When using an UltraWave logical instrument (such as an MWNoiseSource), any vector that contains a POP microcode must be fully defined across all defined logical instrument instances.
  - IG-XL returns an error if you do not follow this guideline.
  - Reapplying the same Signal to a given logical instrument instance results in the hardware NOPing the application. (This prevents glitches.)

For more information and an example, see Using UltraWave24 Instruments with POP.

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UltraWave24 MWNoiseSource Programming : MWNoiseSource Pattern Control : MWNoiseSource ASCII Pattern Syntax

MWNoiseSource ASCII Pattern Syntax

MWNoiseSource pattern microcodes are instructions that you can place on an individual vector of a pattern. For that vector, a microcode instructs the MWNoiseSource to perform a particular action, such as starting a new signal.

You can only use MWNoiseSource microcodes with pins that the instruments statement has specified as MWNoiseSource. For example:

```
instruments = {
        DUT_AIN:MWNoiseSource;
}
```

For more information, see Instruments Statement in the Pattern Language section.

If a vector contains any MWNoiseSource related microcode other than nop, that vector is placed in SRM.

The format for specifying an MWNoiseSource microcode on a particular vector is:

(pin-list : MWNoiseSource = microcode)

The following code shows an example of ASCII pattern microcode syntax for an MWNoiseSource. The figure shows this pattern in PatternTool.

```
instruments = {
        Noise:MWNoiseSource 1;
}

vm_vector
vm_MWNoiseSourceOnly_A ( $tset       , Dig4)
{
                    > BasicSource2  0    ;
```

```
start_label MWNoiseSourceOnly:
                > BasicSource2  0   ;
(Noise:MWNoiseSource = LoadSettingsConnect Hot)
                > BasicSource2  0   ;
repeat 10000        > BasicSource2  0   ; //1ms wait time
(Noise:MWNoiseSource = LoadSettingsConnect Cold)
                > BasicSource2  0   ;
repeat 10000        > BasicSource2  0   ; //1ms wait time
(Noise:MWNoiseSource = Disconnect)
                > BasicSource2  0   ;
repeat 2000         > BasicSource2  0   ; //200us wait time
                > BasicSource2  0   ;
                > BasicSource2  0   ;
                > BasicSource2  0   ;
                > BasicSource2  0   ;
                > BasicSource2  0   ;
                > BasicSource2  0   ;
                > BasicSource2  0   ;
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                > BasicSource2  0   ;
```

```
                    > BasicSource2  0    ;
                    > BasicSource2  0    ;
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                    > BasicSource2  0    ;
halt                 > BasicSource2  0    ;
}
```

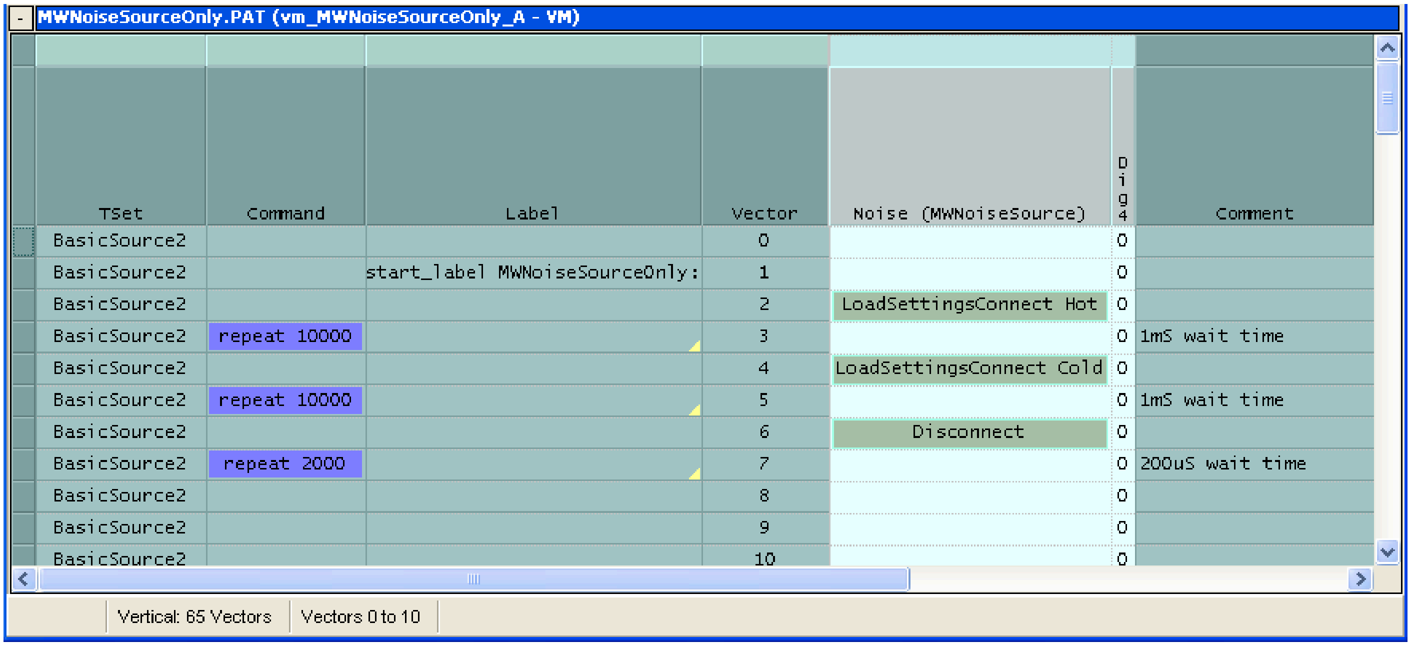

MWNoiseSourceOnly.PAT (vm_MWNoiseSourceOnly_A - VM)

TSet	Command	Label	Vector	Noise (MWNoiseSource)	Dig4	Comment
BasicSource2			0		0	
BasicSource2		start_label MWNoiseSourceOnly:	1		0	
BasicSource2			2	LoadSettingsConnect Hot	0	
BasicSource2	repeat 10000		3		0	1mS wait time
BasicSource2			4	LoadSettingsConnect Cold	0	
BasicSource2	repeat 10000		5		0	1mS wait time
BasicSource2			6	Disconnect	0	
BasicSource2	repeat 2000		7		0	200uS wait time
BasicSource2			8		0	
BasicSource2			9		0	
BasicSource2			10		0	

Vertical: 65 Vectors | Vectors 0 to 10

MWNoiseSource Pattern Microcodes Example

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UltraWave24 MWNoiseSource Theory of Operation

UltraWave24 MWNoiseSource Theory of Operation

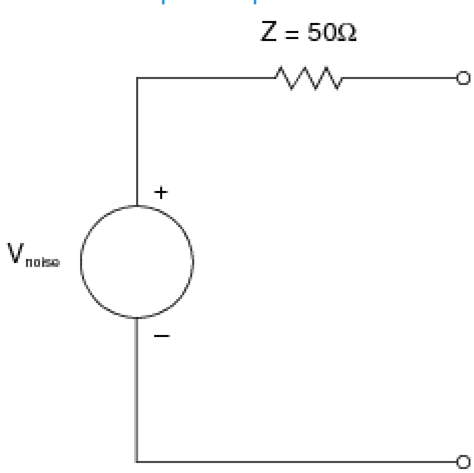
The MWNoiseSource instrument produces a noise signal with a calibrated ENR value. There is a noise source on each of the four front ends of an MW Measure HD board.

This section includes the following topic:

For information on MWNoiseSource instrument performance specifications and standards, see UltraWave24 Specifications.

Noise Source Model

The MWNoiseSource can be modeled as an ideal broadband (white) noise source in series with a 50ohm output impedance. See the figure.



Equivalent Circuit Model of the MWNoiseSource

The noise source supports the Y-factor method for measuring the noise added by a device. The Y-factor method requires a noise source with a calibrated ENR. An accurate ENR value for the MWNoiseSource is calculated and stored during calibration.

The measurement technique based on the Y-factor method requires two measurements:

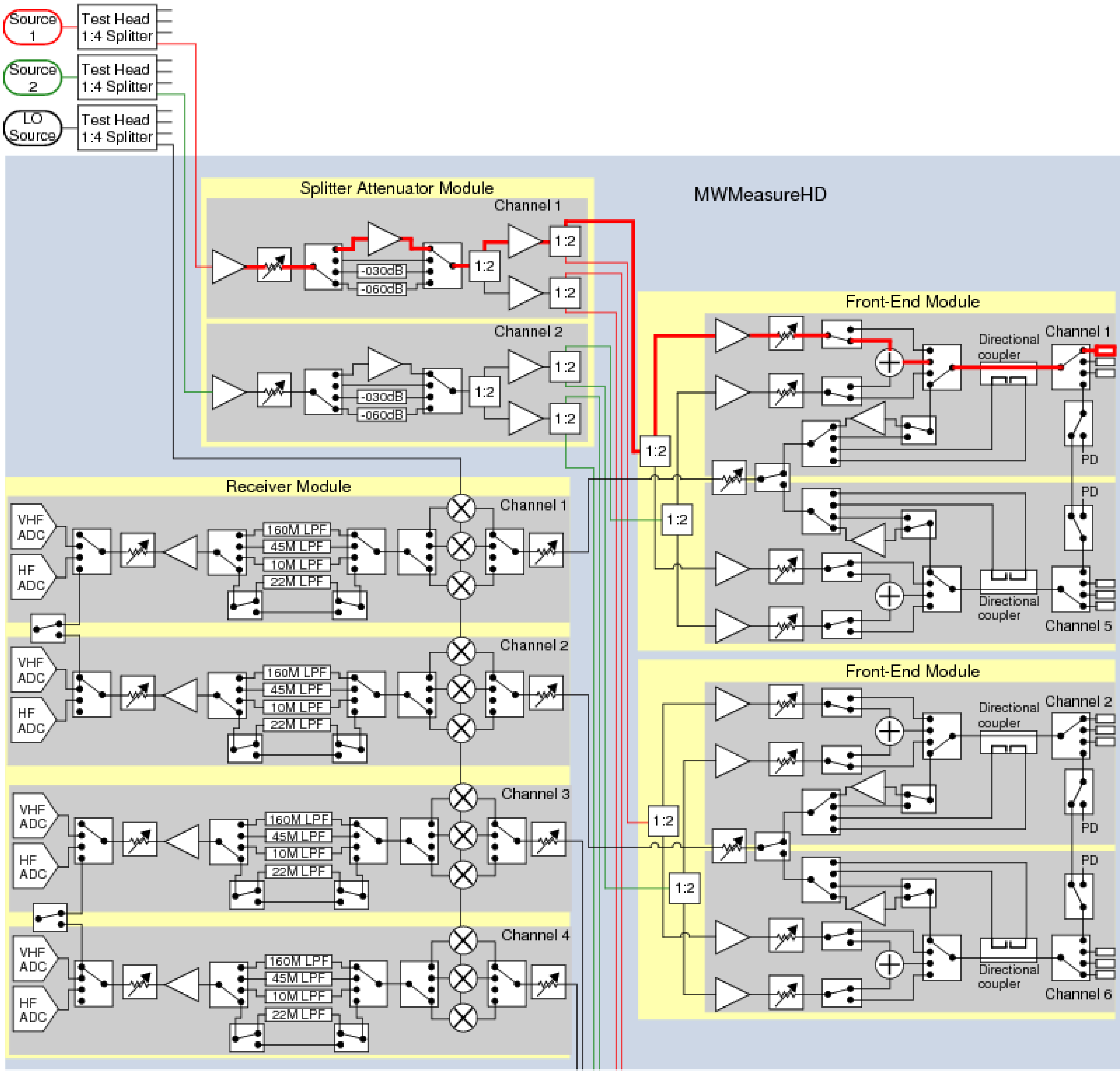
- Noise power for the cold condition

- Noise power for the hot condition

When the noise source is off, it simulates a cold temperature condition. When the noise source is switched on, it simulates a hot temperature condition.

MWNoiseSource Hardware Functions and Signal Flow

The figure shows noise source and signal path to a port. Noise sources are not shared. Heavy lines represent signal flow. See UltraWave24 Hardware Functions for more information about the functions in the diagram.



MWNoiseSource Hardware Functions and Signal Flow, Two of Four FEMs Shown

For DIB connections (MW ports, Pogo pins) and signal names for the MW subsystem and instruments, see User View of UltraWave24 SMPM Connector Blocks, User View of UltraWave24 DIB Pad Patterns, and UltraWave24 Signal Descriptions.

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